

# Kratik Rath

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## WORK EXPERIENCE

- Data Scientist at Project 990 Inc. | Chicago, IL****October 2025 - Present**
  - Processed and analyzed 4.5M IRS tax-return records using Python, SQL and Alteryx to build a nonprofit network and generate grant-making / grant-seeking recommendations.
  - Built a Tableau geographic map of 400K U.S. nonprofits to help grant makers identify and target high-need areas.
  - Migrated 16.5M IRS grant records across 7 years into a production database by engineering an ETL migration pipeline in Python, ensuring zero duplicate ingestion and full audit traceability through exception logging and per-year summary reports.
- Software Engineer Intern at Genmark AI Inc. | Remote, USA****July 2025 - August 2025**
  - Developed the initial version of Gemini-powered RAG pipeline using Python and LangChain, embedding crawled data from GCP Cloud Bucket into a Milvus vector database and containerizing the end-to-end system with Docker to support scalable, context-aware retrieval.
  - Enabled approximately 60% faster content generation for clients by implementing a chat-based interface over embedded assets, producing accurate context-aware responses using Gemini and a Dockerized vector store.
  - Reduced content creation time to under 5 minutes by building a tool that analyzed top-20 trending TikTok videos, generated transcripts via Whisper and produced personalized videos using Google Veo2; recognized as "Most Innovative Project".
- Research Assistant at Indiana University – Indiana Innocence Project | Bloomington, IN****January 2025 - May 2025**
  - Constructed a Python-based ingestion pipeline using BeautifulSoup to scrape and normalize legal case data from intake forms and judicial sources into a PostgreSQL database, cutting manual data entry time from 30 minutes to 1 minute.
  - Designed a Looker Studio dashboard to visualize case timelines, severity and procedural status, helping assign review priority to 200+ cases.
- Data Analyst Intern at Swastika Investmart Ltd. | Indore, India****January 2024 - July 2024**
  - Created an automated analytics workflow using Python, Flask API, SQL and cron jobs to generate daily reports on 50K+ records, providing product and campaign level insights across client zones, which led to a 20% increase in conversions and reduced manual effort by 90%.
  - Engineered a scalable Docker-based scheduler to centralize deployment of internal APIs and integrated Prometheus with unified logging, reducing debugging time by 60% and streamlining future rollouts.
- Data Analyst Intern at Mahindra and Mahindra Ltd. | Mumbai, India****January 2023 - April 2023**
  - Implemented a time-series forecasting model using ARIMAX and SARIMAX to predict financial trends for the next 6 years with 87% accuracy using past 20 years of monthly microeconomic data, which helped forecast sales volume across multiple car models.
  - Compiled and processed a 30K+ record dataset across 20+ car models to analyze part cross-utilization, supporting cost optimization efforts.

## PROJECTS

- California Live Wildfire Analysis Dashboard****December 2025**
  - Architected a Calfire ETL pipeline on AWS Lambda, S3, and BigQuery completing end-to-end in 2.5 seconds, continuously ingesting and refreshing wildfire incident data for dashboard-ready analytics.
  - Built a live Tableau wildfire analytics dashboard with YoY acres burned, YoY incident count KPIs, plus county and seasonal breakdowns to accelerate identification of high-impact regions and trend shifts.
- IndyGo – Ridership Dashboard (Client Project, ENGR-E 583: Information Visualization)****May 2025**
  - Ensured ridership accuracy by validating 38K GPS and APC records, eliminating 3% invalid entries through trip and route-specific thresholds.
  - Designed an interactive Power BI dashboard analyzing 35K+ valid records, surfacing weekday vs weekend travel patterns, stop-level passenger flows and route-level ridership KPIs using DAX calculations, which led to a 3% improvement in service planning.
- DocuVerse – ChatBot****January 2025**
  - Developed a RAG-based document processing app with LangChain and Hugging Face embeddings to automate text chunking and semantic representation, reducing manual document review time by 80%.
  - Enhanced retrieval with FAISS and integrated Gemma2-9b-it to enable context-aware Q&A and structured summarization. Implemented persistent chat history and deployed the system on Streamlit for real-time usage.

## EDUCATION

- Indiana University – Bloomington****August 2024 - May 2026**

Master of Science in Data Science  
GPA: 3.6
- Medi-Caps University****August 2019 - July 2023**

Bachelor of Technology in Computer Science and Engineering  
GPA: 3.4

## TECHNICAL SKILLS

**Programming Languages:** Python, R, SQL  
**Databases & Warehousing:** Snowflake, BigQuery, MySQL, PostgreSQL, MS SQL Server, MongoDB  
**Cloud Platforms & Big Data Technologies:** Databricks, Apache Spark (PySpark), AWS, GCP  
**ETL & Data Visualization:** Alteryx, Power BI (DAX, Power Query), Tableau, Looker Studio, Excel  
**Libraries & Frameworks:** Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Keras, OpenCV, LangChain, Matplotlib, Seaborn, Streamlit, Flask  
**Generative AI & LLMs:** RAG, NLP, FAISS, Milvus, Whisper, Hugging Face  
**Machine Learning:** Regression Models, Tree-based Models (XGBoost, Random Forest), SVM, Clustering (K-Means, GMM), Deep Learning (CNN), Feature Engineering, Statistical Modeling  
**Development Tools:** Git (GitHub, GitLab), Docker, CI/CD, REST APIs, Postman

## CERTIFICATIONS

AWS Certified Cloud Practitioner (January 2026)